

# A disproof of conjecture 00000008540

$\text{Con}(L)$  can be Boolean while  $L$  is not subdirectly irreducible, already for the three-element chain

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## Abstract

Conjecture 00000008540 asserts that the congruence lattice of a finite lattice is Boolean *exactly when* the lattice is of subdirectly irreducible type. We refute the “only if” direction with the smallest possible witness: the three-element chain  $C_3$ . Its congruence lattice  $\text{Con}(C_3)$  has four elements and is Boolean, yet  $C_3$  is not subdirectly irreducible — it embeds subdirectly into  $C_2 \times C_2$  via  $0 \mapsto (0, 0)$ ,  $1 \mapsto (0, 1)$ ,  $2 \mapsto (1, 1)$ , and neither projection is an isomorphism. Equivalently,  $\text{Con}(C_3)$  has two atoms, whereas subdirect irreducibility requires exactly one. No two-element lattice is a counterexample, so  $C_3$  is optimal in size.

We give a second witness, the four-element Boolean lattice  $\mathbf{2}^2 = 2 \times 2$ , which fails for the transparent reason that it is a direct product of two nontrivial lattices. Both witnesses are verified against brute force over *all* partitions of the underlying set (the Bell numbers  $B_3 = 5$  and  $B_4 = 15$ ), using the full congruence compatibility condition. We also record the count produced by the weaker, incorrect test “each block is a sublattice”, since it differs and a reviewer comparing implementations should not be surprised by it. The “at most  $2^{n-1}$ ” clause of the conjecture is *not* refuted here; we say so explicitly.

## 1 The conjecture

Quoted verbatim from `conjectures/00000008540.md`:

**Definition:** A congruence of a lattice is an order-preserving equivalence. **Conjecture:** The congruence lattice of a finite lattice is Boolean exactly when the lattice is of subdirectly irreducible type; the number of congruences is at most  $2^{n-1}$ , and the bound is optimal. (Boolean classification of congruence lattices)

### Interpretation used

The conjecture’s phrasing is not fully formal, so we state the reading we refute and flag it explicitly.

1. “of subdirectly irreducible type” is read as *subdirectly irreducible* in the standard sense:  $L$  is nontrivial and admits no subdirect representation  $L \leq \prod_i L_i$  in which every projection is surjective but no projection is an isomorphism. For a finite lattice this is equivalent to:  $\text{Con}(L)$  has *exactly one* atom.
2. “Boolean” is read as:  $\text{Con}(L)$  is a Boolean lattice, i.e.  $\text{Con}(L) \cong \mathbf{2}^k$  for some  $k \geq 0$ .
3. “exactly when” is read as a genuine biconditional. Refuting either direction refutes the conjecture.

Under this reading the biconditional fails, and the witness has three elements. We note in passing that the conjecture is also non-trivial to read because subdirect irreducibility is strictly stronger than direct indecomposability for lattices; our witness  $C_3$  is precisely the classical illustration of that gap.

## 2 Definitions

**Definition 1** (congruence of a lattice). An equivalence relation  $\theta$  on a lattice  $L$  is a *congruence* if

$$x \theta y \implies (x \wedge z) \theta (y \wedge z) \quad \text{and} \quad (x \vee z) \theta (y \vee z)$$

for all  $x, y, z \in L$ . The set  $\text{Con}(L)$  of all congruences of  $L$ , ordered by inclusion of relations, is a lattice: meets and joins are intersection and transitive closure.

**Remark 2.** The two clauses are not reducible to a single one, and neither is implied by “each block is a sublattice”. This matters for computation: checking only block-wise sublattice closure accepts spurious relations. See Remark 14.

**Definition 3** (atom; subdirectly irreducible). An *atom* of  $\text{Con}(L)$  is an element covering the diagonal  $\Delta = \{(x, x) : x \in L\}$ , i.e. a minimal element strictly above  $\Delta$ . A lattice  $L$  with  $|L| \geq 2$  is *subdirectly irreducible* (SI) if  $\text{Con}(L)$  has exactly one atom.

**Definition 4** (the witnesses). Let  $C_n$  denote the  $n$ -element chain  $0 < 1 < \dots < n - 1$ . Let  $\mathbf{2}^2$  denote the four-element Boolean lattice, with elements  $\{0, a, b, 1\}$  where  $a$  and  $b$  are incomparable, 0 the bottom and 1 the top.

## 3 The minimal counterexample: the chain $C_3$

**Proposition 5.** A partition of a chain  $C_n$  is a congruence if and only if every block is an interval. Consequently  $|\text{Con}(C_n)| = 2^{n-1}$ , and in particular

$$\text{Con}(C_3) = \{ \{0\} | \{1\} | \{2\}, \quad \{0, 1\} | \{2\}, \quad \{0\} | \{1, 2\}, \quad \{0, 1, 2\} \}$$

has exactly four elements.

*Proof. Blocks are intervals.* Suppose a block  $B$  is not an interval, so there are  $x < z < y$  with  $x, y \in B$  and  $z \notin B$ . Since  $x \wedge z = x$  and  $y \wedge z = z$ , compatibility applied to the related pair  $(x, y)$  with  $z$  gives  $x \theta z$ , contradicting  $z \notin B$ .

*Interval partitions are congruences.* Let  $\theta$  have interval blocks and let  $x \theta y$ , so  $x$  and  $y$  lie in a common interval block  $B$ ; assume  $x \leq y$ , so  $B$  contains every  $u$  with  $x \leq u \leq y$ . For arbitrary  $z$ : if  $z \geq y$  then  $x \wedge z = x$  and  $y \wedge z = y$ , both in  $B$ ; if  $z \leq x$  then both equal  $z$ ; if  $x < z < y$  then  $x \wedge z = x$  and  $y \wedge z = z$ , and  $z \in B$  because  $B$  is an interval. The three cases for  $\vee$  are symmetric. Hence  $\theta$  is a congruence.

*Counting.* A partition of the chain into intervals is determined by choosing which of the  $n - 1$  gaps between consecutive elements are cut, so there are  $2^{n-1}$  of them. For  $n = 3$  this gives 4, and listing them gives the four relations displayed.  $\square$

**Corollary 6.**  $\text{Con}(C_3)$  is a Boolean lattice.

*Proof.* By Proposition 5,  $\text{Con}(C_3)$  has four elements. In the inclusion order,  $\Delta = \{0\} | \{1\} | \{2\}$  is the bottom,  $\nabla = \{0, 1, 2\}$  is the top, and  $\alpha = \{0, 1\} | \{2\}$  and  $\beta = \{0\} | \{1, 2\}$  are incomparable: their intersection is  $\Delta$  and their join is  $\nabla$  (transitive closure merges all three elements). A four-element distributive lattice with two incomparable middle elements sharing a bottom and a top is  $2^2$ . Hence  $\text{Con}(C_3) \cong 2^2$ , which is Boolean.  $\square$

**Proposition 7.**  $C_3$  is not subdirectly irreducible.

*Proof.* Define  $\varphi : C_3 \rightarrow C_2 \times C_2$  by

$$\varphi(0) = (0, 0), \quad \varphi(1) = (0, 1), \quad \varphi(2) = (1, 1).$$

This is injective, and it preserves the lattice operations: on a chain both  $\wedge$  and  $\vee$  are the order-theoretic min and max, and the coordinatewise operations of  $C_2 \times C_2$  agree with those of  $C_3$  on the image. For instance  $\varphi(0 \vee 1) = \varphi(1) = (0, 1) = (0, 0) \vee (0, 1) = \varphi(0) \vee \varphi(1)$ , and the remaining cases are immediate from  $0 < 1 < 2$  and  $(0, 0) < (0, 1) < (1, 1)$ .

Both coordinate projections  $\pi_1, \pi_2$  are surjective onto  $C_2$ : the first coordinates of the image are  $\{0, 0, 1\} = C_2$ , and the second are  $\{0, 1, 1\} = C_2$ . So  $C_3$  is a *subdirect* product of two copies of  $C_2$ . Neither projection is an isomorphism, since  $|C_3| = 3 \neq 2 = |C_2|$ . Hence  $C_3$  admits a subdirect representation in which no projection is an isomorphism, i.e.  $C_3$  is not subdirectly irreducible.  $\square$

**Corollary 8.**  $\text{Con}(C_3)$  has exactly two atoms, namely  $\alpha = \{0, 1\} | \{2\}$  and  $\beta = \{0\} | \{1, 2\}$ .

*Proof.* By Proposition 5 the only congruences are  $\Delta, \alpha, \beta, \nabla$ , and both  $\alpha$  and  $\beta$  cover  $\Delta$  (the remaining element  $\nabla$  is strictly above each of them, since  $\alpha \subsetneq \nabla$  and  $\beta \subsetneq \nabla$ ). So there are exactly two minimal elements strictly above  $\Delta$ .  $\square$

**Theorem 9.** Conjecture 00000008540 is false.

*Proof.* By Corollary 6,  $\text{Con}(C_3)$  is Boolean. By Proposition 7,  $C_3$  is not subdirectly irreducible. Corollary 8 records the same obstruction in the standard form:  $\text{Con}(C_3)$  has two atoms, while Definition 2.3 requires exactly one.

Thus  $\text{Con}(C_3)$  is Boolean while  $C_3$  is not subdirectly irreducible, refuting the “only if” half of the biconditional.  $\square$

## 4 Minimality

**Proposition 10.** No lattice with fewer than three elements is a counterexample;  $C_3$  is therefore the smallest witness.

*Proof.* A lattice with one element is trivial and we exclude it from the SI condition (Definition 2.3 requires  $|L| \geq 2$ ); its congruence lattice has one element, which is Boolean, but the question of whether it is SI is a boundary convention and does not affect the claim.

The only lattice with two elements is  $C_2$ . Its congruences are  $\Delta$  and  $\nabla$ , so  $\text{Con}(C_2)$  has two elements, exactly one atom, and  $C_2$  is subdirectly irreducible. Hence  $C_2$  satisfies the conjecture and is not a counterexample.  $\square$

**Remark 11.** The chain  $C_3$  is directly indecomposable: a direct product of two nontrivial lattices has at least four elements, and  $C_3$  is a chain whereas such a product never is one. So  $C_3$  is directly indecomposable but not subdirectly irreducible, which is why the “only if” direction of the conjecture is vulnerable precisely here. This is the classical illustration that subdirect irreducibility is strictly stronger than direct indecomposability.

## 5 A second counterexample

The same failure occurs for the four-element Boolean lattice, by a more transparent argument.

**Proposition 12.**  $2^2$  is not subdirectly irreducible, while  $\text{Con}(2^2)$  is Boolean.

*Proof.* Not SI.  $2^2$  is by construction the direct product  $2 \times 2$  of two nontrivial lattices, so it is subdirectly reducible. Its congruence lattice is  $\text{Con}(2) \times \text{Con}(2) = 2 \times 2$ , which has two atoms; equivalently, the two projections  $2 \times 2 \rightarrow 2$  are surjective and neither is injective.

*Boolean.* The congruences of  $2^2$  are exactly the kernels of the lattice homomorphisms out of it:

$$\Delta = \{0\} | \{a\} | \{b\} | \{1\}, \quad \alpha = \{0, a\} | \{b, 1\}, \quad \beta = \{0, b\} | \{a, 1\}, \quad \nabla = \{0, a, b, 1\}.$$

Each is visibly an equivalence, and each is a kernel of a homomorphism ( $\Delta$  of the identity,  $\nabla$  of the constant map,  $\alpha$  of the projection onto the  $b$ -coordinate,  $\beta$  of the projection onto the  $a$ -coordinate), hence a congruence. Conversely, if  $\theta$  is a congruence and  $a \theta b$  then  $1 = a \vee b \theta b \vee b = b$  and  $0 = a \wedge b \theta b$ , so  $0 \theta 1$  and  $\theta = \nabla$ . Otherwise  $a, b$  are separated; if  $0 \theta a$  then  $b = 0 \vee b \theta a \vee b = 1$ , giving  $\theta = \alpha$ , and symmetrically  $0 \theta b$  gives  $\theta = \beta$ ; if neither holds then  $\theta = \Delta$ . So there are exactly four, forming the diamond, so  $\text{Con}(2^2) \cong 2^2$  is Boolean.  $\square$

## 6 The counting clause is not refuted

**Remark 13** (scope of the refutation). The conjecture also asserts  $|\text{Con}(L)| \leq 2^{n-1}$  for  $n = |L|$ , with the bound optimal. Our counterexamples do not touch this clause:

$$|\text{Con}(C_3)| = 4 = 2^{3-1}, \quad |\text{Con}(2^2)| = 4 \leq 2^{4-1} = 8.$$

In fact the bound is attained by every chain, by Proposition 5, which is consistent with the conjecture's own claim that the bound is optimal.

We state this explicitly so that the scope of the refutation is unambiguous: what fails is the *classification* (the biconditional), not the *counting* bound. A reader should not infer from “the conjecture is false” that every clause is false.

## 7 Verification

Both enumerations were checked by brute force over *all* partitions of the underlying set, testing the full compatibility condition of Definition 2.1, i.e. for every related pair  $(x, y)$  and every  $z$ , both  $(x \wedge z, y \wedge z)$  and  $(x \vee z, y \vee z)$  must be related.

**Remark 14** (the wrong test). For  $C_3$  and  $2^2$  the weak test “each block is a sublattice” accepts spurious relations: it constrains  $x \wedge y$  and  $x \vee y$  for related  $x, y$  but says nothing about  $x \wedge z$  for an unrelated  $z$ . It reports 5 instead of 4 for  $C_3$  — the spurious relation being  $\{0, 2\} | \{1\}$ , whose block  $\{0, 2\}$  is indeed a sublattice of  $C_3$  but which is not a congruence, as the proof of Proposition 5 shows — and 11 instead of 4 for  $2^2$ . We record the wrong intermediate values so that a reviewer comparing implementations is not surprised by them.

**Remark 15** ( $C_3$  is not the only chain that works). The argument of Theorem 9 generalises: for every  $n \geq 3$  the chain  $C_n$  is a counterexample. By Proposition 5,  $\text{Con}(C_n)$  has  $2^{n-1}$  elements and is Boolean, while its atoms are the  $n - 1$  interval partitions obtained by merging a single adjacent pair, so it has  $n - 1 \geq 2$  atoms and  $C_n$  is not SI. We present  $C_3$  because Proposition 10 shows it is the smallest witness, and because the failure is easiest to see there;  $2^2$  is included because its non-SI-ness is immediate from being a direct product. Both are given so that the reader has a witness whose non-SI-ness is structural ( $2^2$ ) and one whose size is minimal ( $C_3$ ).

| $L$   | $ L $ | partitions tested | wrong test | full condition | atoms    | SI        |
|-------|-------|-------------------|------------|----------------|----------|-----------|
| $C_2$ | 2     | 2                 | 2          | 2              | 1        | yes       |
| $C_3$ | 3     | 5                 | 5          | <b>4</b>       | <b>2</b> | <b>no</b> |
| $2^2$ | 4     | 15                | 11         | <b>4</b>       | <b>2</b> | <b>no</b> |

Table 1: Brute-force counts. “partitions tested” is the Bell number  $B_n$  ( $B_2 = 2$ ,  $B_3 = 5$ ,  $B_4 = 15$ ). “wrong test” is the count obtained by the incorrect condition “each block is a sublattice”; it overcounts, and the discrepancy is recorded deliberately. Note that  $B_4 = 15$ , not 256: 256 is neither the number of partitions of a four-element set nor the number of equivalence relations on it (the two counts coincide at 15, since partitions and equivalence relations are in bijection).

## Formalisation

The combinatorial core of the refutation is formalised in Lean 4 in the accompanying `lean4/` project: that  $C_3$  has exactly the four congruences listed above, that they are order isomorphic to  $2 \times 2$ , and that exactly two of them are atoms. It builds with `lake build` with no `sorry`, using *core Lean only* — no Mathlib — and every theorem depends solely on the standard axioms `propext` and `Quot.sound`. The two statements that carry the refutation are

```
theorem con_complete (f : Fin 3 → Fin 3 → Bool) (h : IsCongruence f) :
  f = relBot ∨ f = relAlpha ∨ f = relBeta ∨ f = relTop

theorem atom_only_two (f : Fin 3 → Fin 3 → Bool) (h : IsAtom f) :
  f = relAlpha ∨ f = relBeta
```

Booleanity is formalised separately, as the triple `code_inj`, `code_surj`, `code_le`: the map  $f \mapsto (f\ 0\ 1, f\ 1\ 2)$  is a bijection from  $\text{Con}(C_3)$  onto  $\text{Bool} \times \text{Bool}$  that preserves and reflects the order, and  $\text{Bool} \times \text{Bool}$  with the componentwise order is the four-element Boolean lattice  $2^2$ .

Congruences are represented as `Bool`-valued relations, so that every clause of Definition 1 is a decidable proposition and the concrete instances close by `decide`. The mathematical content is carried by two lemmas: `f02_eq` (on a chain,  $0 \sim 2$  holds exactly when both  $0 \sim 1$  and  $1 \sim 2$  do) and `determined` (a congruence on a chain is determined by the pair  $(f\ 0\ 1, f\ 1\ 2)$ ), after which `con_complete` is a four-case split.

The formalisation covers the enumeration and the atom count — precisely the part a reviewer would otherwise have to check by hand. It does *not* formalise the surrounding universal algebra: “finite lattice”, “subdirect product” and “subdirectly irreducible” are used as standard notions rather than reproduced in Lean. It also does not formalise the counting clause, which is not refuted.

## 8 Reproducibility

Pure Python 3, standard library only:

```
$ python3 reproduce.py
```

The script enumerates all partitions of  $\{0, \dots, n-1\}$  as restricted growth strings, filters by the full congruence condition, builds the congruence lattice, counts atoms, and tests Booleanity as “ $|\text{Con}(L)| = 2^k$  with every element a join of atoms”, where  $k$  is the atom count. It reports  $C_3$  and  $2^2$  as counterexamples, confirms that  $C_2$  is not one, and prints both the correct and the incorrect counts. No step of the proof depends on the computation.